

Exercise 5.1 — Blackjack Value Function Analysis

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How Blackjack Works (Quick Primer)

If you have never played blackjack, here is all you need to know for this exercise.

The goal. Get a hand of cards whose values add up as close to 21 as possible, *without going over*. If your total exceeds 21 you “bust” and lose immediately.

Card values. Number cards (2–10) are worth their face value. Face cards (Jack, Queen, King) are each worth 10. An Ace is special: it can count as either 1 or 11, whichever helps you more.

A round of play. Both you (the player) and the dealer start with two cards. You can see one of the dealer’s cards (the “showing” card) but not the other. On your turn you have two choices:

- **Hit** — take another card (adds to your total, but risks going over 21).
- **Stick** (also called “stand”) — stop taking cards and keep your current total.

After you stick, the dealer plays by a fixed rule: keep hitting until reaching 17 or higher, then stick. Whoever is closer to 21 without busting wins.

Rewards. Win = +1, Lose = −1, Draw = 0. There is no discounting ($\gamma = 1$), so the reward from the end of the game *is* the return for every state visited during that game.

What is a “usable ace”? If you hold an Ace that you are currently counting as 11 (and your total is still ≤ 21), that Ace is called *usable*. It acts like a safety net: if a future card would push you over 21, you can “downgrade” the Ace from 11 to 1 and avoid busting. Once downgraded, the Ace is no longer usable.

The Policy Being Evaluated

The textbook evaluates a very simple (and not very good) policy:

If your hand totals 20 or 21, stick. Otherwise, hit.

This means the player keeps requesting cards until reaching 20 or 21 — risky, because every extra card could push the total over 21.

What the 3D plots show. Each surface plots the *value function* $V(s)$: how good, on average, is it to be in a particular situation? The axes are:

- **Player sum** (12–21) — your current card total.
- **Dealer showing** (Ace through 10) — the one dealer card you can see.
- **Height** (−1 to +1) — the average outcome from that situation. Positive means you tend to win; negative means you tend to lose.

There are two separate surfaces: one for hands *with* a usable ace, one *without*.

Problem Statement

Exercise 5.1 asks us to look at the 500,000-episode plots (the right-hand column of Figure 5.1) and explain three visual features:

1. Why does the surface jump *up* sharply for the last two rows in the rear (player sums 20 and 21)?
2. Why does the surface drop *down* along the entire left edge (dealer showing an Ace)?
3. Why are the front values (low player sums) *higher* in the usable-ace plot than in the no-usable-ace plot?

Reproduced Figure

The figure below was generated by simulating 10,000 and 500,000 blackjack games under the stick-on-20-or-21 policy and recording the average outcome for each state (first-visit Monte Carlo prediction).

Answer 1: The big jump at player sums 20 and 21

What is happening in the plot. The surface is mostly flat and negative (around -0.6 to -0.8) for player sums 12 through 19, then it suddenly shoots up to positive territory at sums 20 and 21.

Why. This jump happens because the policy *changes its behavior* right at sum 20. Think of it in two contrasting scenarios.

Player sum is 19 or below — the policy says “hit.” The player is forced to take another card. With a sum of 19, almost any card except an Ace or a 2 will push the total over 21 (a bust). Specifically, only 3 out of every 13 cards are safe; the other 10 cause an instant loss. So the average outcome from sum 19 is very bad (about -0.75).

Player sum is 20 — the policy says “stick.” Now the player stops and keeps their 20. To beat a 20, the dealer must end up with exactly 21. That is hard to do. So the player wins most of these games, and the average outcome jumps to roughly $+0.15$ to $+0.45$ depending on what the dealer is showing.

The table below shows this cliff from the simulation (no usable ace case):

Player Sum	Avg. outcome vs. Dealer Ace	Avg. outcome vs. Dealer 10
19	-0.779	-0.746
20	$+0.148$	$+0.436$
21	$+0.640$	$+0.888$

Table 1: The jump from sum 19 to sum 20 is nearly a full point — from almost-always-losing to usually-winning — because the policy switches from hit to stick.

In short: sums 20 and 21 are the only ones where the player actually *stops and keeps a strong hand*. Everything below 20 is dragged down by the constant risk of busting while hitting.

Exercise 5.1: Blackjack Value Function (stick on 20/21)

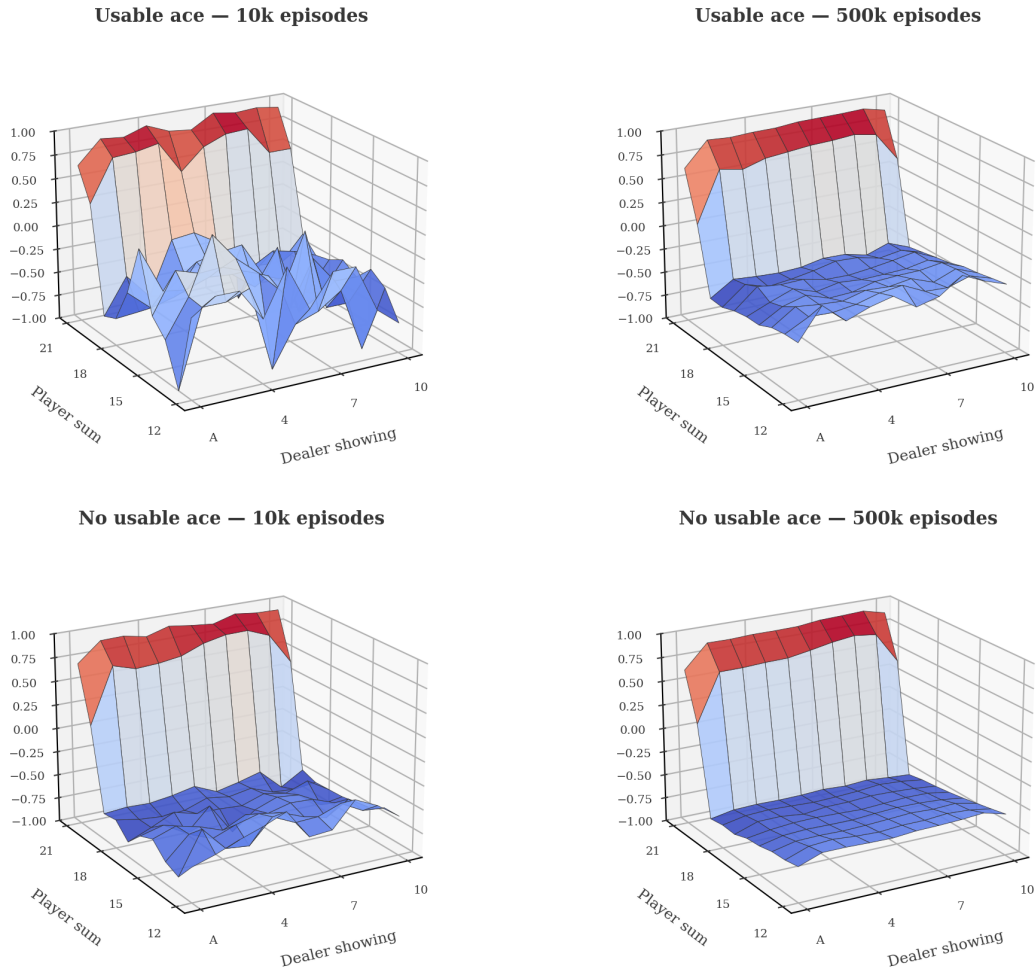


Figure 1: Approximate state-value functions for the stick-on-20-or-21 policy. The 10,000-episode surfaces (left) are noisy; the 500,000-episode surfaces (right) are smooth and accurate.

Answer 2: The drop when the dealer shows an Ace

What is happening in the plot. Along the left edge of each surface (where the dealer's showing card is an Ace), the values are noticeably lower than everywhere else.

Why. An Ace is the best possible card for the dealer, for an intuitive reason: *flexibility*.

Imagine the dealer flips over a 6 as their hidden card, giving them Ace + 6. They can count the Ace as 11 for a total of 17 and stick (a decent hand). Now imagine instead the hidden card is a 7, giving Ace + 7 = 18. Even better. And if a later hit would push the total over 21, the dealer just “downgrades” the Ace from 11 to 1 and keeps going — the Ace acts like a built-in safety net for the dealer too.

Because of this flexibility, the dealer almost never busts when starting with an Ace. The bust rate for a dealer showing an Ace is only about 11.5%. Compare that to a dealer showing a 5 or 6, where the bust rate is roughly 40%. When the dealer busts, the player wins automatically — so a low dealer bust rate means fewer free wins for the player.

In short: the dealer’s Ace makes the dealer’s hand very hard to beat, pulling the player’s average outcome down across *all* player sums.

Answer 3: Usable ace helps at low sums

What is happening in the plot. At the front of the surfaces (low player sums like 12, 13, 14), the *usable ace* plot sits noticeably higher than the *no usable ace* plot.

Why. Remember, under this policy the player hits on everything below 20. At low sums the player will have to hit *multiple times*, and each hit risks a bust. Having a usable ace is like wearing a seatbelt during this process.

Here is a concrete example. Suppose the player’s cards are Ace + 4, totaling 15 (counting the Ace as 11). The player hits and draws a 9. Normally $15 + 9 = 24$, which is a bust. But because the Ace is usable, it downgrades from 11 to 1, changing the total from 24 to 14. The player survives and gets to keep hitting.

Now consider the same situation *without* a usable ace: the player has, say, $8 + 7 = 15$. They hit and draw a 9. The total is $15 + 9 = 24$ — bust, instant loss, -1 reward. No safety net.

This difference shows up clearly in the numbers. At player sum 12 against a dealer showing 10:

$$V(\text{with usable ace}) = -0.279 \tag{1}$$

$$V(\text{without usable ace}) = -0.567 \tag{2}$$

With a usable ace the player still loses on average (negative value), but *much* less often, because the ace absorbs hits that would otherwise cause a bust.

As the player sum gets closer to 20, this advantage shrinks — there is less room for the ace to help, and at sums 20–21 the player sticks immediately regardless, so the usable ace barely matters.

In short: a usable ace is a “get out of bust free” card. It matters most at low sums where the player has to hit many times, making those states significantly less bad.

Summary

Feature	Plain-English Explanation
Jump at sums 20–21	These are the only sums where the player stops hitting and keeps a strong hand. Everything below 20 is terrible because the player is forced to keep drawing cards and often goes over 21.
Drop at dealer Ace	The Ace is the dealer’s best card because it is flexible (can count as 1 or 11). This means the dealer almost never busts, so the player gets fewer “free wins.”
Higher front values with usable ace	A usable ace lets the player survive hits that would otherwise cause a bust — it is a safety net. This matters most at low sums where the player has to hit many times before reaching 20.